CS491 Senior Design Project

Project Specification Document

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1 Introduction

1.1 Description

Sproutly is a mobile application that aims to help individuals who seek to improve their productivity by integrating a wide range of features, such as habit tracking, smart notifications, AI-assisted reminders, task scheduling etc. all in one app. The app is planned to include gamification elements, which may consist of: badges, streaks, or virtual companions; to motivate users to keep their streaks going.

The primary features that differentiate Sproutly from existing applications on the market is diverse feature set, AI integration and ease of use. Many applications aimed at improving productivity have a limited feature set; they may only have a timer, or only have an app locking mechanism, and so on. Sproutly is planned to implement all relevant features in one centralized app, making the user experience more seamless.

Integration of AI also helps make the user experience seamless, by taking care of some of the overhead that may come with setting up schedules, prioritizing certain tasks over others, or determining work/break intervals for a given day. Sproutly aims to adapt a 'it just works' approach, with a streamlined user setup process in the beginning, with the app doing as much of the background work as possible, whilst still offering customizability for users looking to tailor the app further to suit their preferences. Taking care of overhead also comes with the benefit of making the app feel less like another task that has to be done, or another tool that has to be managed, and more like a seamless companion running in the background.

1.2 High Level System Architecture & Components of Proposed Solution

The high-level system architecture can be represented by a range of components, which come together to

- User Interface:
  The user interface is built mobile-first, to make use from a smartphone easy. Flutter will be used to make porting from iOS to Android easier and faster, whilst still keeping as many UI elements and layouts common between the two operating systems as possible.

- AI Module:
  OpenAI, or other alternative AI service providers which offer API keys, will be used. This system is at the core of Sproutly's AI integration that drives its AI-assisted features. Native AI solutions (such as Apple Intelligence) or other similar locally running AI models may also be used.

- Task Scheduling & Task breakdown:
  Users will be able to manually set tasks with deadlines. AI will automatically assist with scheduling these tasks and making recommendations on which task to work on next. The AI will also be able to break down the task into smaller steps. AI is planned to be able to learn from the user's past task scheduling tendencies and give tailored recommendations. Users will have the option to override AI recommendations and breakdowns if they wish.

- Timer:
  Users will be able to set work and break timers, such as pomodoro [1], increasing productivity by managing time more efficiently. AI is planned to learn from past user experience and adjust timer intervals accordingly. Users will have the option to override AI timer recommendations if they wish.

- Proof-of-Work Module:
  At the end of a study or work session, users will have the option to upload a photo as a proof of their work. AI is planned to review and validate user work, helping progress the user's milestone markers.

- App Restriction Module:
  Instead of complete strict app blocking, nudges are planned to be used. If certain apps (such as social media) are detected to be open and running, notifications may be sent upon the user opening these apps, or after a certain period of time has elapsed after opening an app. Overriding the app restriction in harsh and extreme modes will be permitted to users for emergencies with a penalty. The details of the modes are given in the functional requirements part in the report.

- Virtual Companion & Milestones:
  Users will be able to view their progress in a dedicated page. They will be able to view their virtual companion, as well as their productivity streak and any badges or medals they have earned in their journey so far. Badges will be awarded to the user at certain milestones.

- Backend Architecture:
  FastAPI (python) is planned to be used for backend architecture. It will be responsible for making API calls and will act as a mediator between the database and the user frontend.

- Local & Cloud Database:
  Primarily, cloud based MySQL will be used to store user information. Encryption will be used where possible to ensure user data stays protected.

1.3 Constraints

The proposed system is subject to several constraints that are grouped into three categories: implementation constraints, economic constraints and ethical constraints. Implementation constraints describe the technical boundaries of the chosen stack and what is realistically feasible within the project. Economic constraints capture the limits imposed by the project budget, such as the need to rely on free or low-cost services and to control paid API usage. Ethical constraints highlight the boundaries set by privacy, user well-being and responsible use of AI, which restrict how far focus-lock, data collection and gamification mechanisms can go. Together, these constraints define the practical and responsible scope of the project.

1.3.1 Implementation Constraints

The system will be developed as a cross-platform mobile app using Flutter, working for both IOS and Android platforms which constraints the UI design and feature set to what is feasible within Flutter [2]. The backend will be implemented using FastAPI (Python) and hosted on a cloud platform, Railway [3], [4]. This introduces constraints related to RESTful API design. The system will use Railway's managed MySQL database instead of a device-only local database. User data will be stored in the cloud (encrypted and managed by the provider), so users do not lose their data if their phone breaks or if the local app data is deleted. However, this also means the app depends on stable network connectivity and the performance and availability limits of Railway's database service [4]. AI features such as task breakdown, adaptive suggestions and companion dialogue depend on the OpenAI API, so latency, rate limits and a limited budget restrict how frequently the service can be called and how complex each AI interaction can be [5]. Animations will use Rive assets with a limited sprite set, which limits the complexity of the virtual world and storyline to the team's art and design capacity [6].

1.3.2 Economic Constraints

The project is developed under a limited student budget, so the system must use free or low-cost services. Flutter, FastAPI and MySQL are open-source tools and do not require licence fees [2], [3], [7]. Application hosting and the managed database are planned to run on Railway's free or low-price tiers, which keeps monthly costs low but also limits storage and compute power [4]. AI-enabled features rely on the paid OpenAI API. For this project, a small monthly budget is reserved for API usage, and the exact cost depends on the number and size of requests because the service is charged on a per-token basis [5]. Visual assets are planned with Rive, which offers free or affordable plans that are sufficient for a small sprite set [6]. Overall, the technology stack focuses on open-source and low-cost options to control expenses, while still being enough to implement the features defined in the current project scope.

1.3.3 Ethical Constraints

Sproutly is intentionally constrained by several ethical boundaries. It is designed as a supportive focus tool rather than a strict blocking system, so any focus-lock behaviour is

active for a limited period of time, includes an emergency way to exit, and does not take permanent control of the device. To protect privacy, the system does not inspect the content of documents, messages or websites and does not record keystrokes or screenshots; it stores only the minimum information needed for focus statistics, tasks, streaks and optional proof-of-work. User data is not sold or shared with third parties other than the infrastructure and AI providers required to operate the service. AI-generated messages and suggestions are constrained to coaching and are not presented as professional mental-health or medical advice. Streak, reward and notification features are limited and carefully worded so that they do not shame users, push them to work excessively long hours or create an unhealthy dependence on the app.

1.4 Professional and Ethical Issues

The team aims to act in line with professional and academic integrity principles. Project work, code and documentation is produced by the group members and any ideas obtained from external sources will be used only when permitted by their licences and will be properly cited in the report or in code comments. Within the team, tasks related to both documentation and implementation are divided in a fair way so that each member contributes meaningfully and shares the overall project workload. In addition, communication with the supervisor and instructors is kept honest and timely.

1.5 Standards

The development and documentation of Sproutly are planned with reference to the commonly used software engineering standards and guidelines. For requirements and this specification document, which is a standard itself, the structure and use of functional and non-functional requirements follow the general style of the IEEE 830 Software Requirements Specification, adapted to the scope of a student project. For implementation, the backend exposes a RESTful HTTP API using JSON, following widely used web API conventions. Python code written with FastAPI follows the PEP 8 style guide. Basic security recommendations from OWASP are used as a reference when handling authentication and sensitive data.

2 Design Requirements

2.1 Functional Requirements

User Registration and Profile: Users can register and log in via email or Google accounts. They can edit their profile information whenever they want, so they can change their goals or personal details. If they would like to delete their account, they can do that as well. They can also choose to have avatar profile pictures.

Personality Analysis Questionnaire: After signing up, users take a personality test, which consists of Likert-scale questions (such as "strongly agree" to "strongly disagree") and

open-ended questions about their habits and goals. The user can choose how detailed they want the test to be - the quick version is like 10 questions, the detailed version can go up to 50+ questions with situational scenarios like "what do you do when a deadline is tomorrow but you're tired?" The app uses their answers to understand their personality type, work patterns, and stress triggers, which then helps personalize everything from task suggestions to how their companion talks to them.

Coaching Modes: The strictness of the app varies with different modes such as Casual, Standard, Harsh, and Nightmare. In Casual mode, the user will receive soft nudges, and they can override app blocks after 10-15 seconds with minimal penalty. The Standard mode requires a 30-60 second wait to override, and it tracks their streaks. Harsh mode locks the apps as well, also allowing overriding but with a cost, like a few days of cooldown before allowing the user to level up their companion, and significant penalties, such as their companion losing three levels or shutting them out for days, and Nightmare mode makes it even more difficult for the user. For instance, trying to uninstall the app during a focus session triggers massive penalties, and their companion can reset completely to level 1. On iOS, the locking app feature is not allowed; thus, the penalties are made to be harsher than those of Android's. For example, taking the companion's level back to level 1 happens in Harsh mode rather than Nightmare mode.

App Blocking: When the user starts a focus session, the app locks certain distracting apps of the user's choice. On Android, it's a proper lock, but on iOS, it resembles persistent reminders. The blocking strength depends on which coaching mode the user has picked.

Focus Timer: The app includes various timer options, like the classic Pomodoro timer (25 minutes) or custom timers that the user can set themselves. The app can also suggest a timer based on their task. Every time they accomplish a task, their companion grows and the storyline progresses.

Companion System: The user gets their very own virtual pet, such as a cat or donkey. This pet levels up when they stay focused and finish tasks. The companion has its own storyline that has different chapters based on the user's choices and how consistent the user is. If they slack on their work, their companion gets sad or loses progression; however, if they stay focused, it evolves and unlocks new story paths. In nightmare mode, their companion might even die, which means the companion has to go back to level 1.

Task Management: Users can create tasks with deadlines, tags, and subtasks for bigger projects. The app automatically puts the most critical tasks at the top based on their energy levels, deadlines, personality analysis from the questionnaire, and past behavior. This feature stops them from wasting time on less important tasks first and allows them to manage their time better.

Proof of Work: After completing a task, the user has to upload a photo or document as proof so that AI can check whether the user actually completed their task or not. The stricter the

coaching mode the user has chosen before starting their focus session, the harsher AI will be while checking the proof.

AI Task Breakdown: If a task seems to be too big or overwhelming, the app uses AI to split it into smaller steps. The user can either choose the app to break down the task for them, or AI might recommend it beforehand when a user enters a big task into the system. The app takes their task description and creates a clear list of what to do first, second, third, etc. This feature helps users notice that the task is more manageable.

Weekly Progress Review: At the end of each week, the app generates a comprehensive, well-documented report showing the user's focus time, completion rates, and productivity patterns. It includes graphs and charts so that they can see what days or times they work better. The AI also provides personalized feedback on what to improve.

Distraction Inbox: During focus sessions, if random thoughts pop into the user's head, they can quickly jot them down in a distraction inbox. Afterwards, the AI sorts these notes into tasks, reminders, or just notes. This feature helps the user not to break their concentration and helps users to remember their early thoughts.

Mood Check-ins: The app asks how the user is feeling and their energy level throughout the day. Based on their mood, it suggests easier or harder tasks to do at that moment and additionally, shows motivational quotes. This helps them work with their energy instead of facing frustration by trying to do a more challenging task, for which the user has no adequate energy.

Notes Section: There's a built-in notes area for writing down ideas, class notes, or random brain dumps. There is also a scratchpad, which is a temporary space where the user can write anything and later turn pieces into real tasks if needed. The app can automatically organize the user's notes by topic or by date, and AI will make suggestions of possible connections between notes and their existing tasks. The users can search their notes across all their metadata and keywords that the app can identify. Moreover, they can search related content that they may have accidentally overlooked long ago. If they're taking notes in class or meetings, they can use templates that have been pre-made for each subject, or create their own. Their notes would also synchronize across all their devices, so they could start writing on their phone and finish on their laptop. Additionally, the AI can summarize really long notes into bullet points to use as a guide for a quick review, prior to an exam or meeting.

Uploading Schedules: The user can set up their weekly schedule by typing it manually or uploading a screenshot or photo of an existing schedule. The AI reads the image and converts it into a structured calendar that the user can edit. Once their schedule is in the app, it automatically allocates time for their classes, meetings, or any recurring commitments. The app can also detect scheduling conflicts; for instance, if the user has accidentally booked two events at the same time, the app warns them to fix it. If they have a recurring schedule, they can save it as a template so they don't have to re-enter every recurring event each semester or

term. The schedule integrates with the user's task deadlines, so the app knows when they're free actually to work on their tasks.

Home Screen Widgets: The app will be using widgets, which will take place on the user's home screen, showing their companion and giving quick suggestions on what to do next. These tips change throughout the day based on time, deadlines, and their usual habits. It aims to keep the user motivated without opening the app constantly. The user can select from varying widget sizes depending on how much info they want to see: a small-sized widget can show the companion's mood and current streak, whereas a medium one adds their next task and remaining focus time for the day, and a large one can display their full schedule with progress bars. However, this feature differs depending on whether the user is on Android or IOS. If they're on Android, tapping different parts of the widget can take them straight to specific features, such as starting a timer or checking their tasks. The widget also celebrates the wins of the user by showing little animations when the user completes tasks or hits milestones.

2.2 Non-Functional Requirements

2.2.1 Usability

- The application should offer an intuitive, clean, and easy-to-use interface that can be used daily.
- The user experience should remain consistent across iOS and Android platforms, ensuring that users can use the app easily on any device while respecting each platform’s native patterns.
- All core features, like starting a focus session, adding tasks, or checking stats, should be within quick access and require a few steps.
- The interface should stay visually clear and pleasant, with readable fonts, suitable text sizes, and relaxing colors for concentration.
- The design should be smooth and reduce cognitive load by excluding unnecessary elements and presenting information in an organized, simple manner.
- The app should provide helpful onboarding so that new users can understand coaching modes, tasks, the companion, and other core features quickly.

2.2.2 Reliability

- The app should function predictably and consistently, especially during focus sessions and task tracking.
- User progress, streaks, and statistics should remain accurate and consistent across all app activities.
- The app should remain stable without unexpected crashes and handle errors gracefully without interrupting core workflows.

2.2.3 Performance

- The app should run smoothly on standard iOS and Android devices without lag.
- Key features (focus timer, task creation, note-taking, mood check-ins) should load quickly to avoid disrupting user workflow.
- AI-based features such as weekly reports or task suggestions should return results within a reasonable time to maintain a good user experience.
- Animations (such as the companion or transitions) should feel responsive and not significantly impact performance.
- Background syncing operations should not slow down overall app usage.

2.2.4 Supportability

- The system should be easy to maintain, update, and extend over time without requiring major structural changes.
- Code should follow clean, modular architecture to allow quick identification, modification, and addition of features such as new coaching modes or analytics tools.
- The app should include clear documentation (well-documented APIs, code comments, developer guides) for core components (task management, focus sessions, AI features) to support future debugging and improvements.
- Error handling and logging mechanisms should enable efficient issue detection and troubleshooting.
- The design should allow smooth integration of future technologies or external APIs while keeping the existing system stable.

2.2.5 Scalability

- The system should be designed to handle a growing number of users without significant performance decline.
- The data structure should support increasing volumes of notes, tasks, schedules, and focus logs over time.
- The system should support multi-device access, ensuring user data remains consistent across phones and tablets as the number of devices and users grows.
- AI-based features should be designed so they can be improved or expanded in the future as the model evolves.
- The overall architecture should allow adding new coaching modes, companion features, or analytics tools without needing major redesigns.

3 Feasibility Discussions

3.1 Market & Competitive Analysis

3.1.1 Market Overview

Sproutly operates within the productivity, focus, and habit-tracking app ecosystem. Market reports estimate that the habit-tracking market will exceed USD 21 billion by 2031, fueled by smartphone adoption and the mainstreaming of app-based self-improvement tools [8]. More specific studies place the Pomodoro/focus-timer app market at USD 320–412 million in 2024, with projections approaching USD 1 billion by 2033 [9].

As remote and hybrid work environments introduce persistent digital distractions, users increasingly rely on digital self-control tools to regulate attention [10][11]. At the same time, coaching-oriented apps like Fabulous show strong adoption and demonstrate user preference for supportive, behavior-guided systems over simple timers [12]. These broader trends indicate a strong and growing demand for tools like Sproutly that combine productivity support with adaptive guidance.

3.1.2 Competitor Landscape

Several established tools partially overlap with Sproutly's goals:

• Forest - Gamified Focus Timer
– Uses a tree-growing mechanic to discourage phone use [13].
– No personalized task support or adaptive feedback [13].

• Focus To-Do - Pomodoro + Basic Task List
– Combines timers with task lists but lacks intelligent prioritization [14].

• Fabulous - Behavioral Coaching App
– Provides motivational routines based on behavioral science [12].
– Static, predefined journeys; not AI-adaptive [12].

• RescueTime / Freedom - Digital Self-Control Tools
– Focus on usage tracking and blocking [15][16].
– Limited reflective support or personalization [15][16].

These tools address parts of the productivity problem but lack an integrated system that adapts dynamically to the user, assists with large tasks, and provides optional accountability without coercion.

3.1.3 Sproutly's Differentiation & Value Proposition

Sproutly distinguishes itself through:

• AI-mediated focus support, using soft locks and contextual nudges rather than strict blocking.
• Optional proof-of-work, offering accountability only for users who want it.
• AI-driven task breakdown & smart prioritization, helping users manage complex tasks beyond basic timers.
• Lightweight streak-based companion, delivering motivational feedback without the overhead of full coaching systems like Fabulous.

Together, these features position Sproutly as a more adaptive, personalized, and supportive alternative within a crowded timer-dominant market.

3.1.4 Risks & Constraints

Despite strong potential, several risks must be considered:

• High competition & user fatigue
Users already rely on various productivity apps; research shows “tool fatigue” can limit adoption unless a product offers clear value [10]

• Ethical expectations
DSCT tools face scrutiny; persuasive features must remain transparent, optional, and reversible. Sproutly addresses this with opt-in proof-of-work and reversible Focus Lockdown Mode.

• Prototype-stage business limitations
Models relying on LLMs incur operating costs; Sproutly is financially feasible as a senior project but not immediately scalable without a monetization plan.

Overall, Sproutly remains a feasible and competitive prototype, especially for students and knowledge workers seeking a balanced alternative to rigid blocking apps.

3.2 Academic Analysis

Sproutly aligns with current research in Digital Self-Control Tools (DSCTs), digital wellbeing, and persuasive technology. Academic work in these areas provides strong evidence for the feasibility and relevance of Sproutly’s feature set.

3.2.1 Research on Digital Self-Control Tools (DSCTs)

Studies on DSCTs in educational and work settings show that such tools are “numerous and readily available” and often used to manage distraction, especially in digital learning contexts. However, the literature also highlights that many existing tools lack nuanced understanding of how users engage with them in everyday life [11]

Research on distraction-reduction tools indicates that the most effective systems combine timers, blocking options, goal reminders, and self-tracking with reflective features that help users understand and adjust their behaviors, rather than relying only on timers or hard lockdowns [11].

Lyngs et al. further note that the majority of DSCTs follow a limited set of design patterns such as timers, lockouts, countdowns, reminders etc. And emphasize the need for more advanced, theory-driven designs based on Dual Systems theory, which distinguishes between fast impulsive processes (System 1) and slower reflective processes (System 2) [17].

3.2.2 Connection to Sproutly's Feature Set

Sproutly's core features are well-aligned with these findings.

- System 2 (Reflective) support appears through planning tools such as the smart queue, task breakdown, and self-tracking dashboard [17].

- System 1 (Impulsive) support is addressed through soft locks, contextual nudges, and real-time prompts that help users correct distraction immediately [17].

This dual-process alignment follows established cognitive theory on behavior regulation.

3.2.3 Persuasive Technology & Behavior Change

The Fabulous app is widely referenced in persuasive-technology research as a successful habit-formation system. UX reviews and user studies report improvements in routine building, structure, and wellbeing when users consistently engage with supportive, narrative-driven systems [18][19].

Sproutly adopts similar persuasive principles such as goal setting, positive reinforcement, progress visualization, and a motivational companion, but implements them in a lighter, more flexible form suitable for a student project, consistent with guidance from the innovation expert.

3.2.4 Alignment with Academic Recommendations

Sproutly's design closely follows key academic recommendations:

• Support self-regulation rather than coercion

Strict blocking can backfire; DSCT researchers recommend autonomy-supportive interventions [10]. Sproutly implements soft locks, an optional Focus Lockdown Mode, and reversible restrictions, avoiding coercive designs.

• Combine multiple intervention types

Multi-component systems (timers + reminders + tracking + goal-setting) outperform single-feature tools [20] Sproutly integrates focus sessions, smart queue, task breakdown, proof-of-work, mood tracking, and companion feedback.

• Personalization & context-awareness

Research emphasizes personalized interventions that adapt to routines, timing, and user context [11]. Sproutly uses LLMs to tailor focus durations, break tasks down, and provide adaptive guidance.

• Ethical transparency & user control

Design guidelines warn against manipulative, high-pressure systems and encourage opt-in features. Sproutly avoids coercive penalties and uses optional, reversible motivational mechanisms.

4 Glossary

AI: Artificial Intelligence
API: Application Programming Interface
DSCT: Digital Self-Control Tools

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